

Essential Summary

A Generalized Padeé Approximation Method for Solving Homoclinic and Heteroclinic Orbits of Strongly Nonlinear Autonomous Oscillators

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1 Background and Motivation

The classical Padeé approximation—approximating a function by a rational function $P_L(z)/Q_M(z)$ where both numerator and denominator are polynomials—has been widely applied to nonlinear dynamics. Various researchers have extended it by replacing polynomial components with exponential functions or other bases to better capture the asymptotic decay behavior of homoclinic orbits. However, these modifications were proposed on a case-by-case basis, without a **unifying theoretical framework**.

This paper (the English version of the generalized Padeé approximation method) proposes an intrinsic extension: the numerator and denominator of the Padeé approximant are generalized from polynomial functions to **series composed of any kind of function**. This framework unifies all existing modifications theoretically. Furthermore, it extends the method's scope from homoclinic orbits alone to **both homoclinic and heteroclinic orbits**, and validates it on three fundamentally different oscillator classes.

2 Core Innovation

2.1 The Generalized Definition

Let $\hat{H}_m = \{P : P(z) = \sum_{i=0}^m a_i g_i(z)^i, g_i : \mathbb{C} \rightarrow \mathbb{C}, a_i \in \mathbb{C}\}$ and $G(m, n) = \{R = P/Q : P \in \hat{H}_m, Q \in \hat{H}_n \setminus \{0\}\}$. If a rational function $P_L/Q_M \in G(L, M)$ satisfies

$$f(z) - \frac{P_L(z)}{Q_M(z)} = \mathcal{O}(z^{L+M+1}), \quad (1)$$

it is called the **Generalized Padeé Approximant**, denoted $\text{GPA}[L/M]_f$. The classical Padeé and all quasi-Padeé variants (Mikhlin, Manucharyan, Zhang et al.) become special cases of this definition.

2.2 Two Novel Approximants for Homoclinic and Heteroclinic Orbits

For the autonomous oscillator $\ddot{x} + g(x) = 0$ with saddle point $H(h, 0)$, the power series solution $x(t) = \sum_{i=0}^{\infty} a_i t^i$ is first obtained. For homoclinic orbits, the boundary condition $x(\pm\infty) = h$ requires $\lim_{t \rightarrow \pm\infty} x(t) \rightarrow h$, which a power series cannot represent. The GPA method constructs:

For homoclinic orbits (satisfying $x(\pm\infty) = h$):

$$\text{GPA}_{\text{hom}}[L/M] = \frac{\sum_{i=0}^L \alpha_i \text{sech}^i t}{1 + \sum_{i=1}^M \beta_i \text{sech}^i t}. \quad (2)$$

For heteroclinic orbits (satisfying $x(\pm\infty) = h_{1,2}$):

$$\text{GPA}_{\text{het}}[L/M] = \frac{\sum_{i=0}^L \alpha_i \tanh^i t}{1 + \sum_{i=1}^M \beta_i \tanh^i t}. \quad (3)$$

The coefficients $\{\alpha_i, \beta_i\}$ (totaling $L + M + 1$ unknowns) are determined by Taylor-expanding the GPA around $t = 0$ and matching the first $L + M + 1$ coefficients with the power series $\{a_i\}$ —a purely algebraic procedure. The initial condition $a = x(0)$ is obtained from the energy integral $\int_h^a g(x)dx = 0$. For heteroclinic orbits connecting two distinct saddle points h_1 and h_2 , the boundary conditions $x(\pm\infty) = h_{1,2}$ are automatically satisfied by $\tanh^i t$ (limits ± 1 at $\pm\infty$).

2.3 Why Hyperbolic Functions?

The choice of $\text{sech } t$ and $\tanh t$ is motivated by three factors: (1) their asymptotic behavior naturally matches homoclinic/heteroclinic boundary conditions; (2) their Taylor expansions are simpler than exponentials, reducing the computational cost of coefficient matching; and (3) the resulting approximant is a rational function of hyperbolic functions—mathematically compact while capturing the essential asymptotic features.

3 Key Results: Three Oscillator Classes

3.1 Cubic–Quintic Duffing Equation

$$\ddot{x} + c_1 x + c_3 x^3 + c_5 x^5 = 0. \quad (4)$$

With $c_1 = 2, c_3 = -10, c_5 = 9$, the system possesses five fixed points: three centers and two saddles. Two homoclinic orbits (around the outer centers) and a pair of heteroclinic orbits (linking the two saddle points) coexist. GPA[4/4] and GPA[7/7] successfully capture all orbits with phase portraits matching the Runge–Kutta method.

3.2 Duffing Equation with Φ^6 Potential

$$\ddot{x} + c_1 x + c_2 x^2 + c_3 x^3 + c_4 x^4 + c_5 x^5 = 0. \quad (5)$$

This asymmetric five-term polynomial is conservative for the parametrically and externally forced Φ^6 oscillator model. With $c_1 = c_3 = c_5 = 1, c_2 = c_4 = -1.5$, the system has three saddle points. GPA[5/5] and GPA[7/7] capture the homoclinic and heteroclinic orbits, with accuracy demonstrably higher than the classical Padé approximant at the same approximation order.

3.3 Generalized Duffing–Harmonic Oscillator

$$\ddot{x} + \frac{\lambda x + \mu x^3}{1 + \nu x^2} = 0. \quad (6)$$

This rational-restoring-force oscillator serves as a benchmark for approximate methods. When $\lambda = -1, \mu = 5, \nu = 5$, two homoclinic orbits exist; GPA[3/3] captures them. When $\lambda = 3, \mu = -5, \nu = 2$, a pair of heteroclinic orbits are obtained via GPA[8/8]. The rational restoring force—whose exact solution is unknown—poses no additional difficulty for the GPA method.

4 Significance

1. **Unifying framework:** The GPA definition encompasses the classical Padeé approximant and all quasi-Padeé variants (Mikhlin 1995, Manucharyan & Mikhlin 2005, Zhang et al. 2011) as special cases, providing a common theoretical language for the field.
2. **First simultaneous homoclinic–heteroclinic treatment:** The dual approximant construction (sech t for homoclinic, tanh t for heteroclinic) provides a unified computational pipeline for both orbit types—a capability absent from prior Padeé-based methods.
3. **Superior accuracy vs. classical Padeé:** At the same approximation order, the hyperbolic-function GPA achieves demonstrably higher accuracy than polynomial-based Padeé approximants, particularly for strongly nonlinear oscillators where polynomial bases require high orders to approximate exponential decay.

5 Limitations

- The method solves for orbits of conservative ($\varepsilon = 0$) systems only; damped or forced systems require separate treatment.
- The approximation order $L = M$ is chosen empirically; no systematic error bound is available.
- The coefficient-matching procedure involves solving a system of $L + M + 1$ nonlinear algebraic equations, which may become ill-conditioned for very high orders.

6 Relationship to the GHFP/MGHFP Series

The GPA method is methodologically distinct from the GHFP/MGHFP series: GPA constructs homoclinic/heteroclinic solutions directly via rational function approximation in time domain, whereas GHFP/MGHFP uses perturbation theory in angular domain to derive amplitude–parameter relationships. However, both share the common goal of **analytical treatment of strongly nonlinear oscillators** and share a common mathematical toolkit—the use of rational function approximation. The 2013 and 2014 GPA papers introduced hyperbolic-function approximants that later inspired the $x = a \sin^2 \varphi + b$ homoclinic solution construction in the GHFP framework (JSV 2013), which appeared simultaneously with the P7 paper.